Solutions: Chapter 5 In-Class Questions

EMS611U Conceptual Design of Aircraft • Week 8 • Dr Hülya Biler • QMUL

Question 1 Power loading for cruise at altitude

Idea. The engine must deliver, at 16,404 ft, the power that gives 22 lb/hp at the cruise weight. Correct for weight ($W/W_0 = 0.9$) and for the power lapse with altitude (Fig. 5.2):

$$\left(\frac{W}{hp}\right)_{TO} = \left(\frac{W}{hp}\right)_{cruise} \frac{W_0}{W_{cruise}} \frac{hp_{cruise}}{hp_{TO}} = 22 \times \frac{1}{0.9} \times \frac{hp_{16{,}404}}{hp_{SL}}$$

Power ratios read from Fig. 5.2 (straight lines through the plotted end points):

- **IO-360** (normally aspirated): about 200 hp at sea level, falling to about 107 hp at 16,404 ft, so $hp_{16{,}404}/hp_{SL} \approx 0.54$. Check: the usual approximation $P/P_{SL} = 1.132\,\sigma - 0.132$ with $\sigma = 0.601$ at 5,000 m gives 0.55.
- **TO-360** (turbocharged): about 210 hp, held almost constant up to roughly 10,000–11,000 ft, then about 176–181 hp at 16,404 ft, so $hp_{16{,}404}/hp_{SL} \approx 0.85$.

(a) Turbocharged (TO-360):

$$\left(\frac{W}{hp}\right)_{TO} = 22 \times \frac{0.85}{0.9} \approx 20.8 \text{ lb/hp}$$

(b) Not supercharged (IO-360):

$$\left(\frac{W}{hp}\right)_{TO} = 22 \times \frac{0.54}{0.9} \approx 13.2 \text{ lb/hp}$$

(c) The turbocharged engine needs about $20.8/13.2 \approx 1.6$ times less sea-level power for the same aircraft, so it is the smaller, lighter installation for high-altitude cruise, despite the turbocharger's own weight and cost. The normally aspirated engine would be heavily oversized at sea level.

Note. This assumes the engine can run at full available power in cruise. If cruise is flown at the usual 75% of available power, multiply both answers by 0.75: 15.6 lb/hp (turbocharged) and 9.9 lb/hp (normally aspirated). Readings of Fig. 5.2 within about ±5% are acceptable.

Question 2 T/W and W/S for the ASW aircraft

Reference aircraft (approximate public data): Lockheed S-3 Viking: maximum take-off weight about 52,500 lb (normal about 42,500 lb); wing area about 598 ft^2; two GE TF34 turbofans of 9,275 lb each. So $T/W \approx 0.35$ (maximum weight) to 0.44 (normal), and $W/S \approx 88$ to 71 lb/ft^2.

Assumptions: $A = 7$, $e = 0.8$, clean $C_{D_0} = 0.015$; landing $C_{L_{max}} = 2.4$ (Fowler flaps with leading-edge devices), take-off $C_{L_{max}} = 0.8 \times 2.4 = 1.92$; ISA at 30,000 ft: $p = 629.7$ psf, $a = 994.8$ ft/s.

(a) Thrust-to-weight ratio

Method	Working	T/W
Table 5.1	Military cargo/bomber	0.25
Table 5.3, $M_{max} \approx 0.75$	$0.244 \times 0.75^{0.341}$ (cargo/bomber); $0.267 \times 0.75^{0.363}$ (transport)	0.22–0.24
Thrust matching	$(L/D)_{cr} = 0.866 \times 16 = 13.9$; $W_{cr}/W_0 = 0.956$; high-bypass lapse at 30,000 ft ≈ 0.25 (Fig. 5.1): $\frac{1}{13.9} \times 0.956 \times \frac{1}{0.25}$	0.28
One engine out, $G = 0.024$	$C_{D_0} = 0.015 + 0.02$ (take-off flaps), $e = 0.8 \times 0.95$: $0.024 + 2\sqrt{\frac{0.035}{\pi \times 7 \times 0.76}} = 0.116$ per engine, $\times 2$	≥ 0.23

Thrust matching gives the highest value. Adding a margin for carrier wave-off (go-around) and hot days:

$$\boxed{T/W = 0.30}$$

(b) Wing loading for each requirement

Requirement	Working	W/S at condition	$(W/S)_{TO}$
Carrier approach	$V_{approach} = 110$ kt (assumed) $= 1.15\,V_{stall}$, so $V_{stall} = 95.7$ kt $= 161.4$ ft/s; $\frac{1}{2}(0.002377)(161.4)^2(2.4)$; $W_{land}/W_0 \approx 0.70$	74	106
Catapult	$V_{end} = 110$ kt (assumed) $+V_{wod} = 20$ kt $= 219.4$ ft/s; $\frac{1}{2}(0.002377)(219.4)^2 \times 1.92/1.21$	–	91
Cruise (jet range)	$q = 0.7 \times 629.7 \times 0.6^2 = 158.7$ psf; $q\sqrt{\pi \times 7 \times 0.8 \times 0.015/3}$; $\div 0.85$	47	55
Loiter (jet endurance)	10,000 ft, 180 kt (assumed): $q = 81.0$ psf; $q\sqrt{\pi \times 7 \times 0.8 \times 0.015}$; $\div 0.85$	42	49

(c) Selection

The loiter and cruise values (49–55 lb/ft^2) are optima for part of the mission and lie well below Table 5.5 (jet transport/bomber 120 lb/ft^2), so by themselves they could be ignored. For an ASW aircraft, however, the 3-hour loiter is a large part of the mission, so the wing should not stray far from them. The carrier limits (91 and 106 lb/ft^2) must not be exceeded. A compromise well inside the carrier limits and close to the S-3:

$$\boxed{W/S = 70 \text{ lb/ft}^2 \quad \text{and} \quad T/W = 0.30}$$

- Wing area: $S = 56{,}700/70 = 810$ ft^2 (75 m^2), against about 598 ft^2 for the S-3.
- Thrust: $T = 0.30 \times 56{,}700 = 17{,}000$ lb, i.e. about 8,500 lb per engine, against 9,275 lb for the S-3.
- The differences come mainly from weight: our first-pass W_0 (56,700 lb) is well above the S-3's normal take-off weight (42,500 lb).

Checks.

- One engine out: each engine gives $0.30/2 = 0.15 > 0.116$. ✓

- Cruise lift coefficient: $C_L = (70 \times 0.956)/158.7 = 0.42$, close to the typical design C_l of about 0.5 (Week 5). ✓
- Both values are close to, but slightly below, the S-3's (W/S 71–88 lb/ft^2, T/W 0.35–0.44). The S-3's higher T/W reflects carrier wave-off and climb requirements that a later, more detailed analysis would add.

These values now feed the refined sizing of Chapter 6 and must be rechecked once the drag polar is known.